Parametric vs Nonparametric Spectrum Estimation

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Key Concepts

1) Spectrum estimation is the problem of inferring the power spectrum of a random process from measured data.

    a) Nonparametric spectrum estimation does not assume any model for the data.

    b) Parametric spectrum estimation assumes the data is generated by a time series model, such as autoregressive, moving average, or sinusoids in noise.

2) Nonparametric spectrum estimation relies on the discrete Fourier transform.

3) Parametric spectrum estimation uses the data to estimate the parameters of the assumed model and then expresses the power spectrum as a function of the estimated model parameters.

    a) High quality spectrum estimates may be obtained when the assumed model is correct.

    b) If the model is incorrect, wrong and/or misleading spectrum estimates may result.

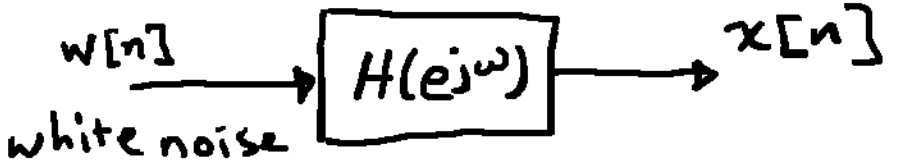
Video Screenshots

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Nonparametric spectrum estimation - no assumed model
- Fourier based using DFT

Parametric spectrum estimation - model based

For example:



$$\Rightarrow S_{xx}(\omega) = S_{ww}(\omega)\,|H(e^{j\omega})|^2$$

$$= \sigma^2\,|H(e^{j\omega})|^2$$

Auto regressive (AR): all-pole
Moving Average (MA): all-zero
Autoregressive Moving Average (ARMA): pole-zero

Sinusoids in noise model

$$x[n] = \sum_{i=1}^{L} A_i e^{j\omega_i n} + w[n] \Rightarrow S_{xx}(\omega) = \sum_{i=1}^{L} \sigma_i^2 \delta(\omega - \omega_i) + \sigma_w^2$$

unknown: $\sigma_i^2, \omega_i, \sigma_w^2$

Parametric Spectrum Estimation

1) Observe $x[n]$

2) Estimate model parameters

- AR, MA, ARMA, $\{\sigma_i^2, \omega_i, \sigma_w^2, i=1,\dots L\}$ ...

3) Express $\hat{S}_{xx}(\omega)$ as a function of model parameters

$$\hat{S}_{xx}(\omega) = \hat{\sigma}^2 |\hat{H}(e^{j\omega})|^2, \quad \hat{S}_{xx}(\omega) = \sum_{i=1}^{L} \hat{\sigma}_i \delta(\omega - \hat{\omega}_i) + \hat{\sigma}_w^2$$

Comparison

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If model is correct -

- High quality spectrum estimates
- Significantly less data required

If model is wrong -

- Parametric can give wrong/misleading estimates

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